Matrices

Introduction and Types of Matrices

Introduction

- A matrix is a **rectangular arrangement** of numbers (or elements) in rows and columns, enclosed in brackets.
- A matrix with m rows and n columns is called an $m \times n$ matrix.
- The element in the i-th row and j-th column is denoted by a_{ij}.

General Form:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Types of Matrices

1 Row Matrix

A matrix that has **one row** $(1 \times n)$.

Example: $A = \begin{bmatrix} 2 & 4 & -1 \end{bmatrix}$

2 Column Matrix

A matrix that has **one column** $(m \times 1)$.

Example: $A = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}$

3 Rectangular Matrix

A matrix with different numbers of rows and columns $(m \times n, m \neq n)$.

Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

4 Square Matrix

A matrix with the **same number** of rows and columns $(n \times n)$.

Example: $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$

5 Zero (Null) Matrix

A matrix in which all elements are zero.

Example: $O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

6 Identity (Unit) Matrix

A **square** matrix with 1's on the main diagonal and 0's elsewhere.

Example: $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

7 Diagonal Matrix

A square matrix in which all off-diagonal elements are zero.

Example: $A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

8 Scalar Matrix

A diagonal matrix in which all diagonal elements are equal to a constant k.

Example: $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ $(k = 3)$

9 Upper Triangular Matrix

A square matrix in which all elements below the main diagonal are zero.

Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$

10 Lower Triangular Matrix

A square matrix in which all elements above the main diagonal are zero.

Example: $A = \begin{bmatrix} 7 & 0 & 0 \\ 2 & 8 & 0 \\ 3 & 1 & 9 \end{bmatrix}$

11 Symmetric Matrix

A square matrix that is equal to its transpose, $A = A^T$.

Example: $A = \begin{bmatrix} 2 & 4 & 1 \\ 4 & 3 & 0 \\ 1 & 0 & 5 \end{bmatrix}$

12 Skew-Symmetric Matrix

A square matrix that satisfies $A^T = -A$ (the diagonal elements are zero).

Example: $A = \begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{bmatrix}$

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Matrices

Equality, Addition, Subtraction and Transpose

Introduction

- Matrices can be **compared**, **added**, **subtracted**, and **transposed**. These operations are defined for
- matrices of compatible dimensions (same order).

Objectives

- Define equality of matrices.
- Perform addition and subtraction of matrices.
- Find the transpose of a matrix.

1 Equality of Matrices

Definition: Two matrices A and B are **equal**, denoted by $A = B$, if they have the **same order** and their corresponding elements are equal, i.e., $a_{ij} = b_{ij}$ for all i, j.

Example: Let $A = \begin{bmatrix} 2 & -1 \\ 5 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 5 & 3 \end{bmatrix}$.

Then $A = B$.

2 Addition of Matrices

Definition: If A and B are two matrices of the **same order**, then their **sum** $A + B$ is obtained by adding the corresponding elements: $(A + B)_{ij} = a_{ij} + b_{ij}$.

Example: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & -1 \\ 2 & 0 \end{bmatrix}$.

$$A + B = \begin{bmatrix} 1+5 & 2+(-1) \\ 3+2 & 4+0 \end{bmatrix} = \begin{bmatrix} 6 & 1 \\ 5 & 4 \end{bmatrix}.$$

3 Subtraction of Matrices

Definition: If A and B are two matrices of the **same order**, then their **difference** $A - B$ is obtained by **subtracting** the corresponding elements: $(A - B)_{ij} = a_{ij} - b_{ij}$.

Example: Let $A = \begin{bmatrix} 4 & 7 \\ 6 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 5 & 1 \end{bmatrix}$.

$$A - B = \begin{bmatrix} 4-1 & 7-2 \\ 6-5 & 3-1 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}.$$

4 Transpose of a Matrix

Definition: The **transpose** of a matrix A, denoted by A^T, is the matrix obtained by interchanging its rows and columns: $(A^T)_{ij} = a_{ji}$.

Example: Let $A = \begin{bmatrix} 2 & 5 \\ 3 & 6 \end{bmatrix}$.

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Matrices

Practice Exercises with Solutions

The following 15 questions cover: **Types of Matrices, Equality of Matrices, Addition, Subtraction, and Transpose.**

Q1 Identify the type of the matrix.

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Solution:

A is a **diagonal matrix** and also a **scalar matrix** (with $k = 3$).

Q2 Classify each matrix.

$$A = \begin{bmatrix} 2 & 4 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

Solution:

A: Row matrix (1×3).

B: Column matrix (3×1).

Q3 Find the order and type.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

Order: 2×3.

Type: **Rectangular matrix** (not square).

Q4 Find x and y.

$$\begin{bmatrix} x & 3 \\ 5 & y \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$$

Solution:

By equality of matrices: $x = 2$, $y = 7$.

Q5 Find a, b and c.

$$\begin{bmatrix} a & 4 & c \\ 2 & b & 6 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 8 \\ 2 & 5 & 6 \end{bmatrix}$$

Solution:

$a = 3$, $b = 5$, $c = 8$.

Q6 Are the matrices equal? Why?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Solution:

Not equal, because: $a_{12} \neq b_{11}$ $(2 \neq 3)$ and $a_{21} \neq b_{21}$ $(3 \neq 2)$.

Q7 Find $A + B$.

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}$$

Solution:

$$A + B = \begin{bmatrix} 2+1 & 3+6 \\ 4+2 & 5+3 \end{bmatrix} = \begin{bmatrix} 3 & 9 \\ 6 & 8 \end{bmatrix}$$

Q8 Compute $A + B$.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$$

Solution:

$$A + B = \begin{bmatrix} 1+6 & 2+5 & 3+4 \\ 4+3 & 5+2 & 6+1 \end{bmatrix} = \begin{bmatrix} 7 & 7 & 7 \\ 7 & 7 & 7 \end{bmatrix}$$

Q9 Can the matrices be added? Explain.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Solution:

No. The orders are 2×3 and 2×2, so addition is not defined.

Q10 Find $A - B$.

$$A = \begin{bmatrix} 8 & 7 \\ 6 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

Solution:

$$A - B = \begin{bmatrix} 8-3 & 7-2 \\ 6-1 & 5-4 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 5 & 1 \end{bmatrix}$$

Q11 Compute $A - B$.

$$A = \begin{bmatrix} -2 & 5 & 7 \\ 4 & 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -1 & 2 \\ -2 & 6 & 1 \end{bmatrix}$$

Solution:

$$A - B = \begin{bmatrix} -2-3 & 5-(-1) & 7-2 \\ 4-(-2) & 0-6 & 3-1 \end{bmatrix} = \begin{bmatrix} -5 & 6 & 5 \\ 6 & -6 & 2 \end{bmatrix}$$

Q12 Is subtraction possible? Explain.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

No. The orders are 3×2 and 2×3, so subtraction is not defined.

Q13 Find the transpose of A.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Q14 Find B^T.

$$B = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}$$

Solution:

$$B^T = \begin{bmatrix} 2 & 3 & 6 \\ 0 & 4 & 7 \\ 1 & 5 & 8 \end{bmatrix}$$

Q15 Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$.

(a) Find $A + B$.

(b) Find $(A + B)^T$.

(c) Find $A^T + B^T$.

(d) Are (b) and (c) equal?

Solution:

(a) $A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$

(b) $(A + B)^T = \begin{bmatrix} 6 & 10 \\ 8 & 12 \end{bmatrix}$

(c) $A^T + B^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 10 \\ 8 & 12 \end{bmatrix}$

(d) Yes, $(A + B)^T = A^T + B^T$.

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Matrices – Practice Questions with Solutions

Types – Equality – Addition – Subtraction – Transpose

Q1 Type of Matrix

Identify the type of the matrix:

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Solution:

- A is a square matrix (3×3).
- All off-diagonal elements are 0.
- All diagonal elements are equal (3).

$\therefore$ **A is a scalar matrix.**

Q2 Types of Matrices

Classify each matrix:

$$A = \begin{bmatrix} 2 & 4 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}.$$

Solution:

- A has one row (1×3) → Row matrix.
- B has one column (3×1) → **Column matrix.**

Q3 Order and Type

Find the order and type of the matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

- A has 2 rows and 3 columns.
- Order: 2×3.
- Since rows $\neq$ columns, A is a **rectangular matrix.**

Q4 Equality of Matrices

If $\begin{bmatrix} x & 3 \\ 5 & y \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$, find x and y.

Solution:

By equality of matrices, corresponding elements are equal:

$x = 2, \; y = 7.$

Q5 Find Unknowns

Find a, b, c, given that:

$$\begin{bmatrix} a & 4 & c \\ 2 & b & 6 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 8 \\ 2 & 5 & 6 \end{bmatrix}.$$

Solution:

From equality of corresponding elements:

$a = 3, \; b = 5, \; c = 8.$

Q6 Equality Check

Are A and B equal? Give a reason.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Solution:

Compare corresponding elements:

$a_{12} = 2 \neq 3 = b_{12}$ and $a_{13} = 3 \neq 2 = b_{13}$.

$\therefore$ **$A \neq B$ (not equal).**

Q7 Addition of Matrices

Find $A + B$.

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}$$

Solution:

$$A + B = \begin{bmatrix} 2+1 & 3+6 \\ 4+2 & 5+3 \end{bmatrix} = \begin{bmatrix} 3 & 9 \\ 6 & 8 \end{bmatrix}$$

Q8 Sum of Matrices

Calculate $A + B$.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$$

Solution:

$$A + B = \begin{bmatrix} 1+6 & 2+5 & 3+4 \\ 4+3 & 5+2 & 6+1 \end{bmatrix} = \begin{bmatrix} 7 & 7 & 7 \\ 7 & 7 & 7 \end{bmatrix}$$

Q9 Possibility of Addition

Can the matrices be added? Explain.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Solution:

- A is 2×3.
- B is 2×2.
- The orders are different.

$\therefore$ **Addition is not possible.**

Q10 Subtraction of Matrices

Find $A - B$.

$$A = \begin{bmatrix} 8 & 7 \\ 6 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

Solution:

$$A - B = \begin{bmatrix} 8-3 & 7-2 \\ 6-1 & 5-4 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 5 & 1 \end{bmatrix}$$

Q11 Subtraction

Calculate $A - B$.

$$A = \begin{bmatrix} -2 & 5 & 7 \\ 4 & 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -1 & 2 \\ -2 & 6 & 1 \end{bmatrix}$$

Solution:

$$A - B = \begin{bmatrix} -2-3 & 5-(-1) & 7-2 \\ 4-(-2) & 0-6 & 3-1 \end{bmatrix} = \begin{bmatrix} -5 & 6 & 5 \\ 6 & -6 & 2 \end{bmatrix}$$

Q12 Possibility of Subtraction

Determine whether subtraction is possible.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

- A is 3×2, B is 2×3.
- Different orders.

$\therefore$ **Subtraction is not possible.**

Q13 Transpose of a Matrix

Find the transpose of A.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Q14 Transpose

Find B^T.

$$B = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}$$

Solution:

$$B^T = \begin{bmatrix} 2 & 3 & 6 \\ 0 & 4 & 7 \\ 1 & 5 & 8 \end{bmatrix}$$

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Q15 Mixed Problem

Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$.

(1) Find $A + B$. (2) Find $(A + B)^T$.
(3) Find $A^T + B^T$. (4) Are (2) and (3) equal?

Solution:

(1) $A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$

(2) $(A + B)^T = \begin{bmatrix} 6 & 10 \\ 8 & 12 \end{bmatrix}$

(3) $A^T + B^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 10 \\ 8 & 12 \end{bmatrix}$

(4) **Yes, $(A + B)^T = A^T + B^T$.**

Matrices

Practice Questions with Solutions



- Types of Matrices
- Equality of Matrices
- Addition of Matrices
- Subtraction of Matrices
- Transpose of a Matrix

15 Solved Questions

Q1 – Q3: Types of Matrices

Q1 Identify the type of the matrix:

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Solution:

A is a **diagonal matrix** (special case: scalar matrix with $k = 3$) of order 3×3.

Q2 Classify each matrix:

$$A = \begin{bmatrix} 2 & 4 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

Solution:

A: Row matrix (1×3).
B: Column matrix (3×1).

Q3 Find the order of A and decide whether it is square or rectangular:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

Order: 2×3.
A is a **rectangular matrix** (not square).

Q4 – Q6: Equality of Matrices

Q4 If $\begin{bmatrix} x & 3 \\ 5 & y \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$, find x and y.

Solution:

By equality of matrices: $x = 2$, $y = 7$.

Q5 Find a, b, c, given that

$$\begin{bmatrix} a & 4 & c \\ 2 & b & 6 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 8 \\ 2 & 5 & 6 \end{bmatrix}.$$

Solution:

$a = 3$, $b = 5$, $c = 8$.

Q6 Are the matrices equal?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}.$$

Solution:

No. They are not equal, because the corresponding elements are not the same (e.g., $a_{12} = 2 \neq 3$).

Q7 – Q9: Addition of Matrices

Q7 Find $A + B$, where

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}.$$

Solution:

$$A + B = \begin{bmatrix} 2+1 & 3+6 \\ 4+2 & 5+3 \end{bmatrix} = \begin{bmatrix} 3 & 9 \\ 6 & 8 \end{bmatrix}$$

Q8 Calculate the sum:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}.$$

Solution:

$$A + B = \begin{bmatrix} 7 & 7 & 7 \\ 7 & 7 & 7 \end{bmatrix}.$$

Q9 Can the matrices be added?

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Solution:

No. They cannot be added, because their orders are different (2×3 and 2×2).

Q10 – Q12: Subtraction of Matrices

Q10 Find $A - B$, where

$$A = \begin{bmatrix} 8 & 7 \\ 6 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}.$$

Solution:

$$A - B = \begin{bmatrix} 8-3 & 7-2 \\ 6-1 & 5-4 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 5 & 1 \end{bmatrix}$$

Q11 Calculate $A - B$, where

$$A = \begin{bmatrix} -2 & 5 & 7 \\ 4 & 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -1 & 2 \\ -2 & 6 & 1 \end{bmatrix}.$$

Solution:

$$A + B = \begin{bmatrix} -2-3 & 5-(-1) & 7-2 \\ 4-(-2) & 0-6 & 3-1 \end{bmatrix} = \begin{bmatrix} -5 & 6 & 5 \\ 6 & -6 & 2 \end{bmatrix}$$

Q12 Is subtraction possible?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

No. Subtraction is not possible because their orders are different (3×2 and 2×3).

Q13 – Q15: Transpose and Mixed Operations

Q13 Find the transpose of A:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Q14 Find B^T, where

$$B = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}.$$

Solution:

$$B^T = \begin{bmatrix} 2 & 3 & 6 \\ 0 & 4 & 7 \\ 1 & 5 & 8 \end{bmatrix}.$$

Q15 Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$, find:

(1) $A + B$ (2) $(A + B)^T$ (3) $A^T + B^T$

Are (2) and (3) equal?

Solution:

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

$$(A + B)^T = \begin{bmatrix} 6 & 10 \\ 8 & 12 \end{bmatrix}$$

$$A^T + B^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 10 \\ 8 & 12 \end{bmatrix}$$

Yes, they are equal since $(A + B)^T = A^T + B^T$.

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Matrices – Practice Questions and Solutions

Types, Equality, Addition, Subtraction and Transpose

Q1 Identify the type of the matrix:

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Solution:

- A is a square matrix (3×3).
- All off-diagonal elements are zero.
- All diagonal elements are equal (3).

Thus, A is a **scalar matrix**.

Q2 Classify each matrix:

$$A = \begin{bmatrix} 2 & 4 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

Solution:

- A has one row (1×3) → **Row matrix**.
- B has one column (3×1) → **Column matrix**.

Q3 State the order and type of:

State the order and type of:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

- Number of rows = 2, number of columns = 3.
- Order = 2×3.
- Since rows ≠ columns, A is a **rectangular matrix**.

Q4 Find the order and type of.

If $\begin{bmatrix} x & 3 \\ 5 & y \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix}$, find x and y.

Solution:

By equality of matrices, corresponding elements are equal:

$x = 2, \quad y = 7.$

$\boxed{x = 2, \quad y = 7}$

Q5 Find a, b, c, given that:

Find a, b, c, given that:

$$A = \begin{bmatrix} a & 4 & c \\ 2 & b & 6 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 8 \\ 2 & 5 & 6 \end{bmatrix}$$

Solution:

- Equating corresponding elements:

$a = 3, \quad b = 5, \quad c = 8.$

$\boxed{a = 3, \quad b = 5, \quad c = 8.}$

Q6 Are the matrices equal?

$A = [1 \ 3], \quad B = [1 \ 3]$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Solution:

- Both are 2×2, but
- $a_{12} = 2 \neq 3 = b_{12}$.
- $\therefore A \neq B$.

$\boxed{A \neq B.}$

Q7 Find A + B:

Find $A + B$:

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}$$

Solution:

$$A + B = \begin{bmatrix} 2+1 & 3+6 \\ 4+2 & 5+3 \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 3 & 9 \\ 6 & 8 \end{bmatrix}}$$

Q8 Calculate the sum:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$$

Solution:

$$A + B = \begin{bmatrix} 1+6 & 2+5 & 3+4 \\ 4+3 & 5+2 & 6+1 \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 7 & 7 & 7 \\ 7 & 7 & 7 \end{bmatrix}}$$

Q9 Can the matrices be added?

Can the matrices be added?

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Solution:

- A is 2×3 and B is 2×2.
- Addition is possible only for matrices of the **same order**.
- $\therefore$ They **cannot be added.**

Q10 Find A − B:

Find $A - B$:

$$A = \begin{bmatrix} 8 & 7 \\ 6 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

Solution:

$$A - B = \begin{bmatrix} 8-3 & 7-2 \\ 6-1 & 5-4 \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} 5 & 5 \\ 5 & 1 \end{bmatrix}}$$

Q11 Calculate A − B:

Calculate $A - B$:

$$A = \begin{bmatrix} -2 & 5 & 7 \\ 4 & 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -1 & 2 \\ -2 & 6 & 1 \end{bmatrix}$$

Solution:

$$A - B = \begin{bmatrix} -2-3 & 5-(-1) & 7-2 \\ 4-(-2) & 0-6 & 3-1 \end{bmatrix}$$

$$= \boxed{\begin{bmatrix} -5 & 6 & 5 \\ 6 & -6 & 2 \end{bmatrix}}$$

Q12 Is subtraction possible?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution:

- A is 3×2 and B is 2×3.
- Subtraction is possible only for matrices of the **same** order.
- $\therefore$ Subtraction is **not possible.**

Q13 Find the transpose of A.

Find the transpose of A:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}.$$

Solution:

- A^T is obtained by interchanging rows and columns:

$$A^T = \boxed{\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}}$$

Q14 Find B^T:

Find B^T:

$$B = \begin{bmatrix} 2 & 0 & 1 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}.$$

Solution:

$$B^T = \boxed{\begin{bmatrix} 2 & 3 & 6 \\ 0 & 4 & 7 \\ 1 & 5 & 8 \end{bmatrix}}$$

Q15 Given A − ...

Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & * \\ 7 & * \end{bmatrix}$, find:

Find: (1) $A + B$ (2) $(A + B)^T$ (3) $A^T + B^T$

Are (2) and (3) equal?

Solution:

1. $A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$
2. $(A + B)^T = \begin{bmatrix} 6 & 10 \\ 8 & 12 \end{bmatrix}$
3. $A^T + B^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix}$

Comparison:

$(A + B)^T = A^T + B^T.$

∴ Yes, they are equal.

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Matrix Multiplication

Definition – Conditions – Examples – Properties – Exercises

Introduction

- Matrix multiplication (**product**) is an operation that combines **two matrices** to produce a new **matrix**.

Objective

- Define the product of two matrices.
- State the condition for multiplication.
- Compute the product of matrices.
- Know the main properties of multiplication.

Main Result

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$.

1 Definition

Let $A = [a_{ij}]$ be an $m \times n$ matrix and $B = [b_{ij}]$ be an $n \times p$ matrix. The product AB is the $m \times p$ matrix $C = [c_{ij}]$, where

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

2 Condition for Multiplication

The product AB is defined only if the number of **columns** of A equals the number of **rows** of B.

If A is $m \times n$ and B is $n \times p$, then AB exists and is $m \times p$.

3 Step-by-Step Example

Example: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$. Find AB.

Solution:

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

4 Properties of Matrix Multiplication

1. **Not Commutative:** $AB \neq BA$ (in general).
2. **Associative:** $(AB)C = A(BC)$.
3. **Distributive over Addition:** $A(B + C) = AB + AC$, $(A + B)C = AC + BC$.
4. **Identity Matrix:** $AI = IA = A$.
5. **Zero Matrix:** $A0 = 0A = 0$.

5 Special Cases

- **Product with Identity:** $A \cdot I_n = I_m \cdot A = A$.
- **Product with Zero:** $A \cdot 0 = 0 \cdot A = 0$.
- **Product of a Row and Column:** $(1 \times n)(n \times 1)$ gives a 1×1 (a scalar). $(n \times 1)(1 \times n)$ gives an $n \times n$ matrix.

6 Practice Exercises

1 Find AB, where $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -1 \\ 5 & 2 \end{bmatrix}$.

Solution:

$$AB = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 5 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \cdot 4 + 1 \cdot 5 & 2 \cdot (-1) + 1 \cdot 2 \\ 0 \cdot 4 + 3 \cdot 5 & 0 \cdot (-1) + 3 \cdot 2 \end{bmatrix}$$

$$= \begin{bmatrix} 13 & 0 \\ 15 & 6 \end{bmatrix}.$$

2 Find AB, where $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 4 & 2 \end{bmatrix}$.

Solution:

$$AB = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 4 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \cdot 2 + 2 \cdot 0 + 0 \cdot 4 & 1 \cdot 1 + 2 \cdot (-1) \\ -1 \cdot 2 + 3 \cdot 0 + 1 \cdot 4 & -1 \cdot 1 + 3 \cdot (-1) \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 \\ 2 & -2 \end{bmatrix}.$$

3 Determine whether the product AB is defined. If it is defined, find its order.

(a) A is 2×3 and B is 3×4.

(b) A is 3×2 and B is 3×3.

Solution:

(a) Defined, because $3 = 3$. The order of AB is 2×4.

(b) Not defined, because the number of columns of A (2) is not equal to the number of rows of B (3).

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Matrix Multiplication

Examples and Exercises with Solutions with Solutions

Definition

- The product of two matrices $A = [a_{ij}]$ $(m \times n)$ and $B = [b_{ij}]$ $(n \times p)$ is the matrix $C = AB = [c_{ij}]$ $(m \times p)$, where

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

Condition for Multiplication

- The product AB is defined only if the number of columns of A equals the number of rows of B.

If A is $m \times n$ and B is $n \times p$, then AB exists and is $m \times p$.

Main Result

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$.

Properties of Multiplication

1. **Not Commutative:** $AB \neq BA$ (in general).
2. **Associative:** $(AB)C = A(BC)$.
3. **Distributive:** $A(B + C) = AB + AC$.
4. **Identity:** $AI = IA = A$.
5. **Zero:** $A0 = 0A = 0$.

Example 1 — Multiplication of Two 2×2 Matrices

Let $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ 5 & 0 \end{bmatrix}$.

Find AB.

Solution:

$$AB = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 & -2 \\ 5 & 0 \end{bmatrix} = \begin{bmatrix} 2\cdot1+1\cdot5 & 2\cdot(-2)+1\cdot0 \\ 3\cdot1+4\cdot5 & 3\cdot(-2)+4\cdot0 \end{bmatrix} = \begin{bmatrix} 7 & -4 \\ 23 & -6 \end{bmatrix}.$$

Example 2 — Multiplication of a 2×3 and a 3×2 Matrix

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & -2 \\ 5 & 3 \end{bmatrix}$.

Find AB.

Solution:

$$AB = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 0 & -2 \\ 5 & 3 \end{bmatrix} = \begin{bmatrix} 1\cdot2+2\cdot0+0\cdot5 & 1\cdot1+2\cdot(-2)+0\cdot3 \\ -1\cdot2+3\cdot0+4\cdot5 & -1\cdot1+3\cdot(-2)+4\cdot3 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 18 & 5 \end{bmatrix}.$$

Example 3 — Multiplication of Two 3×3 Matrices

Let $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 & 0 \\ 0 & -1 & 4 \\ 3 & 2 & 1 \end{bmatrix}$.

Find AB.

Solution:

$$AB = \begin{bmatrix} 1\cdot2+0\cdot0+2\cdot3 & 1\cdot1+0\cdot(-1)+2\cdot2 & 1\cdot0+0\cdot4+2\cdot1 \\ 2\cdot2+1\cdot0+(-1)\cdot3 & 2\cdot1+1\cdot(-1)+(-1)\cdot2 & 2\cdot0+1\cdot4+(-1)\cdot1 \\ 0\cdot2+3\cdot0+1\cdot3 & 0\cdot1+3\cdot(-1)+1\cdot2 & 0\cdot0+3\cdot4+1\cdot1 \end{bmatrix} = \begin{bmatrix} 8 & 5 & 2 \\ 1 & -1 & 3 \\ 3 & -1 & 13 \end{bmatrix}.$$

Practice Exercises (with Solutions)

1 Find AB, where $A = \begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 \\ 5 & -2 \end{bmatrix}$.

Solution:

$$AB = \begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 4 & 1 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 3\cdot4+(-1)\cdot5 & 3\cdot1+(-1)\cdot(-2) \\ 0\cdot4+2\cdot5 & 0\cdot1+2\cdot(-2) \end{bmatrix} = \begin{bmatrix} 7 & 5 \\ 10 & -4 \end{bmatrix}.$$

2 Find AB, where $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \\ 5 & -1 \end{bmatrix}$.

Solution:

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 4 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 1 & 3 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} 1\cdot2+2\cdot1+3\cdot5 & 1\cdot0+2\cdot3+3\cdot(-1) \\ 0\cdot2+(-1)\cdot1+4\cdot5 & 0\cdot0+(-1)\cdot3+4\cdot(-1) \end{bmatrix} = \begin{bmatrix} 19 & 3 \\ 19 & -7 \end{bmatrix}.$$

3 Find A^2, where $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Solution:

$$A^2 = A \cdot A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1\cdot1+2\cdot3 & 1\cdot2+2\cdot4 \\ 3\cdot1+4\cdot3 & 3\cdot2+4\cdot4 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}.$$

4 Verify the distributive law by computing $A(B + C)$ and $AB + AC$.

$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, $C = \begin{bmatrix} -1 & 2 \\ 4 & 0 \end{bmatrix}$.

Solution:

$$B + C = \begin{bmatrix} 1 & 3 \\ 4 & 3 \end{bmatrix}$$

$$A(B + C) = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 6 & 9 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix}$$

$$AC = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$AB + AC = \begin{bmatrix} 2 & 1 \\ 4 & 5 \end{bmatrix} + \begin{bmatrix} -1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 6 & 9 \end{bmatrix}$$

Hence, $A(B + C) = AB + AC$.

5 Determine whether AB is defined. If yes, find its order.

(a) A is 2×3, B is 3×4.
(b) A is 3×2, B is 3×3.

Solution:

(a) Yes, AB is defined because $3 = 3$. The order of AB is 2×4.

(b) No, AB is not defined because the number of columns of A (2) is not equal to the number of rows of B (3).

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Matrix Multiplication

Definition – Rule – Examples – Properties – Exercises

Definition

- The product of two matrices A and B is a new matrix, denoted by AB, obtained by multiplying the **rows** of A by the **columns** of B.

Rule (Row-by-Column)

- $A = [a_{ij}]$ is an $m \times n$ matrix and
- $B = [b_{ij}]$ is an $\times\ p$ matrix, then

$C = AB = [c_{ij}]$ is an $m \times p$ matrix,

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{bj}$$

Condition for Multiplication

- The product AB is defined only if
- the number of columns of A **equals** the number of rows of B.

If A is $m \times n$ and B is $n \times p$, then AB exists and is $m \times p$.

Main Properties

1. **Not commutative:** $AB \neq BA$ (in general).
2. **Associative:** $(AB)C = A(BC)$.
3. **Distributive over addition:** $A(B + C) = AB + AC$, $(A + B)C = AC + BC$.
4. **Identity:** $AI_n = I_m A = A$.
5. **Zero:** $A0 = 0A = 0$.

Worked Examples

Example 1 — Multiplication of Two 2 × 2 Matrices

Let $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$. Find AB.

Solution:

$$AB = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2(1) + 1(-2) & 2(5) + 1(3) \\ 3(1) + 4(-2) & 3(5) + 4(3) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 13 \\ -5 & 27 \end{bmatrix}.$$

Example 2 — Multiplication of 2 × 3 and 3 × 2

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & -3 \\ 5 & 2 \end{bmatrix}$ Find AB.

Solution:

$$AB = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & -3 \\ 5 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1(2) + 2(0) + 0(5) & 1(1) + 2(-3) + 0(2) \\ -1(2) + 3(0) + 4(5) & -1(1) + 3(-3) + 4(2) \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -5 \\ 18 & 0 \end{bmatrix}.$$

Example 3 — Multiplication of Two 3 × 3 Matrices

Let $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 4 \\ 1 & 0 & 2 \end{bmatrix}$ Find AB.

Solution:

$$AB = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 4 \\ 1 & 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1(2) + 0(-1) + 2(1) & 1(1) + 0(3) + 2(0) & 1(0) + 0(4) + 2(2) \\ 2(2) + 1(-1) + (-1)(1) & 2(1) + 1(3) + (-1)(0) & 2(0) + 1(4) + (-1)(2) \\ 0(2) + 3(-1) + 1(1) & 0(1) + 3(3) + 1(0) & 0(0) + 3(4) + 1(2) \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 1 & 4 \\ 2 & 5 & 2 \\ -2 & 9 & 14 \end{bmatrix}$$

Practice Exercises

1 Find AB.

$A = \begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 0 \\ 5 & 2 \end{bmatrix}$

Solution:

$$AB = \begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 5 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 23 & 8 \\ 1 & -2 \end{bmatrix}$$

2 Find AB.

$A = \begin{bmatrix} 2 & -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$

Solution:

$$AB = \begin{bmatrix} 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix}$$

$$= [2(1) + (-1)(4) + 3(-2)]$$

$$= [-8]$$

3 Find AB.

$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 2 & 0 \\ -1 & 4 \end{bmatrix}$

Solution:

$$AB = \begin{bmatrix} 0 & 1 & 2 \\ 1 & -1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 0 \\ -1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0(3) + 1(2) + 2(-1) & 0(1) + 1(0) + 2(4) \\ 1(3) + (-1)(2) + 3(-1) & 1(1) + (-1)(0) + 3(4) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 8 \\ -2 & 13 \end{bmatrix}.$$

4 Find AB.

$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -1 & 3 \\ 2 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & -1 \\ 0 & 2 & 4 \end{bmatrix}$

Solution:

$$AB = \begin{bmatrix} 4 & 2 & 6 \end{bmatrix}$$

5 Determine the order of AB and compute it.

$A = \begin{bmatrix} 1 & 0 \\ -2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 1 & 2 \\ 0 & -1 & 5 \end{bmatrix}$

Solution:

A is 2×2 and B is 2×3, so AB is 2×3.

$$AB = \begin{bmatrix} 4 & 1 & 2 \\ -8 & -5 & 11 \end{bmatrix}.$$

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